

GRADE 6 MATHEMATICS

Classroom Observation Guide

Grade 6 Mathematics — Day Two: “Getting to Know You”

Professor Xavier Honablue, M.Ed.

Prepared for visiting guests, administrators, and instructional observers.

Purpose of This Guide

Thank you for observing our Grade 6 mathematics classroom today. Day One introduced what math is and how our E3 Math Station System works; today's lesson builds directly on that foundation by turning yesterday's actual classroom results into a data lesson, then having students build one of five geometric shapes into a personal survey. This guide explains the structure of today's lesson, the Common Core and Michigan academic standards it addresses, and the educational research behind the instructional choices you will see in action.

Lesson at a Glance (55 Minutes)

Time	Segment	What's Happening
0-4 min	Welcome Back	Students are greeted with a callback to yesterday's energy - several gave the teacher a pound on the way out the door, and one student called it a “W class.”
4-12 min	Let's Look at Our Data	The class learns that 12 of 42 students' work was hung as exceptional. Students see a live bar chart and calculate the fraction ($12/42 = 2/7$) and percent (about 28.6%).
12-18 min	What This Data Means	A growth-mindset discussion: proficiency is a current measurement, not a fixed judgment. The word “yet” is introduced as a framing tool.
18-22 min	Today's Lesson Overview	Students learn that today's Getting-to-Know-You activity uses five different geometric shapes, each its own survey, answered through drawing instead of writing.
22-45 min	Shape Station Work Time	Each table is assigned one of five shapes - hexagon, pentagon, parallelogram, trapezoid, or octagon -

45-50 min	Turn It In	and builds it with a ruler, divides it into sections, and draws an answer to each survey prompt. Completed shape pages are submitted to the class collection box for display.
50-55 min	What You Just Did	Standards are named in student-friendly language; students copy the “I can” statements into their journal and write 2-3 sentences of evidence for each.

The Data Moment: Turning Yesterday's Results Into Today's Lesson

Rather than a hypothetical data set, today's opening lesson uses the class's own real numbers from Day One. Twelve of the class's forty-two students had their work hung on the wall as exceptional. The teacher uses this honestly, without inflating or hiding the number, to teach fractions, percents, and - just as importantly - a growth-mindset framing for the thirty students who are not yet on the wall.

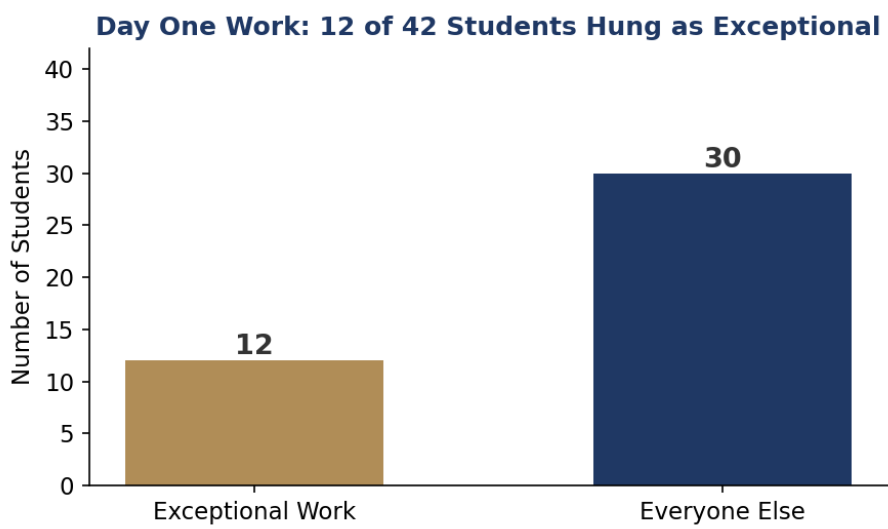


Figure 1. The data visualization shown to students at the start of today's lesson.

The Five Shape Survey Stations

Every station follows the same structure - a grade-6 geometric shape, built with a ruler and divided into sections, with one survey question drawn (not written) per section - but each shape has its own theme and question set.

Shape	Sections	Theme	Sample Prompts
Hexagon	6	All About My Faves	Color, music artist, food, invention, place, weekend

Pentagon	5	My Family & Home	Siblings, birth order, pet, role model, tradition
Parallelogram	4	Who I Am Becoming	Skill to build, hobby, curious subject, source of pride
Trapezoid	3	Quick Takes	One-word self-description, morning/night, current obsession
Octagon ("STOP & Grow")	8	Growth Mindset Survey	A time you tried something hard; what "yet" means to you; a mistake that taught you something

The Octagon station is intentionally shaped like a STOP sign - a familiar 8-sided real-world object - and is entirely dedicated to a positive, growth-mindset survey, extending the data conversation from the start of class into a personal, reflective activity.

Common Core & Michigan Standards Addressed

Mathematics

- **6.G.A.1:** Find the area of shapes composed of triangles, quadrilaterals, and other polygons - the same shapes built at each station.
- **6.G.A.3:** Draw polygons and use coordinates or side lengths to describe them.
- **6.G.A.4:** Represent and reason about 2-D figures, including how they can be divided into equal, symmetrical, or sequential parts.
- **6.SP.B.4:** Display numerical data - used in today's Exceptional Work bar chart.
- **6.SP.B.5:** Summarize a data set in relation to its context - the 12-out-of-42 discussion.
- **6.RP.A.3c:** Find a percent of a quantity - used to turn 12 out of 42 into a percent.
- **MP4, MP5, MP6:** Model with mathematics, use tools strategically, and attend to precision - the same practices from Day One, applied again today.

English Language Arts

- **SL.6.1:** Engage effectively in a range of collaborative discussions - today's data turn-and-talk.
- **W.6.3:** Develop real experiences using effective description - expressed through drawing instead of writing.
- **L.6.1:** Use conventions of standard English when labeling and discussing shapes and sections.

Social-Emotional Learning

The Octagon station and the opening data discussion connect to the CASEL core competencies of self-awareness and self-management, and to Carol Dweck's research on growth mindset - the idea that abilities can be developed through effort, not just fixed traits.

Note: the survey questions themselves (favorite color, siblings, music, etc.) are intentionally not tied to a specific math content standard. They are a vehicle for community-building and ELA/SEL skills, carried inside a math-standard-aligned geometric shape.

Pedagogical & Cognitive Research Foundations

Today's lesson extends the research base introduced on Day One with two additional, specifically relevant frameworks.

Formative Assessment as a Teaching Tool

Opening class with the class's own real performance data - rather than a hypothetical example - reflects Black and Wiliam's (1998) foundational research on formative assessment, which found that using assessment data as feedback to guide instruction and student self-understanding is one of the most effective levers available for raising achievement, particularly for students who are not yet at the top of the class.

Growth Mindset

The explicit framing of proficiency as a “yet,” not a fixed judgment, reflects Carol Dweck's (2006) research distinguishing a growth mindset - the belief that ability can be developed through effort - from a fixed mindset. Dweck's research shows that how a teacher frames results, especially early results, shapes whether students persist or disengage.

Concrete-Representational-Abstract (CRA) Sequence

As on Day One, students continue to build shapes concretely (with a ruler on graph paper) before any symbolic or abstract shape language is introduced, consistent with CRA research (Witzel, Riccomini, & Schneider, 2008).

Gradual Release of Responsibility

The progression from whole-class data discussion, to teacher-modeled shape construction, to independent station work continues the “I do, we do, you do” structure described by Pearson and Gallagher (1983), first introduced on Day One and reinforced today.

What to Look For as an Observer

- How students respond to the data reveal - listen for language shifting from “I'm not good at this” toward “I'm not there yet.”
- The precision students bring to dividing their shape into equal, symmetrical, or sequential sections - notice how the division strategy changes depending on the shape.
- Students choosing to communicate through drawing rather than words - watch how much information a single small drawing can carry.
- The tone at the Octagon “STOP & Grow” station in particular - this is where the data conversation from the start of class becomes personal.

Appendix: Day Two in the Grade 6 Math Lesson Plan Template

For administrators cross-referencing today's observation against the school's required weekly lesson plan format, the table below shows how today's lesson populates that same template.

Template Field	Day Two Content
Standard(s) & Taxonomy Level	6.G.A.1, 6.G.A.3, 6.G.A.4, 6.SP.B.4, 6.SP.B.5, 6.RP.A.3c, MP4-MP6 C/A
Learning Target(s) (SWBAT)	SWBAT describe part of a group using a fraction, decimal,

Success Criteria (I Can)	and percent; construct a grade-6 geometric shape and divide it into equal, symmetrical, or sequential sections; and represent personal information through drawing.
Daily Marzano Strategies	I can use a fraction, decimal, and percent to describe part of a group. I can construct a grade-6 geometric shape and divide it into sections. I can represent information about myself using a drawing instead of words. I can describe what a growth mindset looks like.
Do-Now (5 min)	Identifying Critical Content; Helping Students Examine Their Reasoning; Helping Students Practice Skills, Strategies, and Processes
Instructional Plan	Students are welcomed back with a callback to yesterday's classroom energy before data is revealed.
Critical Content Debrief (5 min)	Students are welcomed back with a callback to yesterday's classroom energy before data is revealed.
Daily Formative [Ungraded] / Summative [Graded] Assessments	Reveal and discuss the 12-of-42 data point using a bar chart, fraction, and percent. Bridge to a growth-mindset framing of proficiency. Introduce the five shape stations and assign each table one shape. Students construct their shape, divide it into sections, and draw an answer to each survey prompt for their assigned shape.
	Can students accurately describe the class data using a fraction and percent, and correctly divide their assigned shape into the required number of sections?
	Formative: teacher observation during the data discussion and station work; journal entries during closing reflection. Summative: none today - completed shape pages are collected as a classroom-community artifact.

Planning to Close the Achievement Gap Using Data

Tier 1 Supports	Tier 2 Supports	Tier 3 Supports
Whole-class modeling of shape construction before station work; the data chart displayed throughout the discussion for visual reference.	Pre-drawn shape outlines with section lines lightly marked for students needing construction support; sentence-free drawing prompts repeated aloud as needed.	One-to-one support during station work; a simplified shape (fewer sections) available for students who need it; extra think time before the growth-mindset discussion.

Data Source: Classroom Formative Assessments [Ungraded]; Day One Summative Work Sample.

Explore More: Resources for Observers

For guests who would like to look deeper into the frameworks behind today's data-and-mindset opening and the shape stations that followed.

About Today's Data Approach

Inside the Black Box (Summary)

A widely cited summary of Black and Wiliam's foundational formative-assessment research, the basis for today's data-driven opening.

[Read the summary](#)

Growth Mindset and Enhanced Learning

A concise overview of Carol Dweck's growth-mindset research and its classroom applications, from Stanford's Teaching Commons.

[Read on Stanford Teaching Commons](#)

For Deeper Research & Standards

Standards for Mathematical Practice

The official Common Core description of MP4, MP5, MP6, and the other five practices referenced throughout today's lesson.

[View on corestandards.org](http://corestandards.org)

CASEL SEL Framework

The Collaborative for Academic, Social, and Emotional Learning's five core competencies, which ground the growth-mindset elements woven into today's Octagon station.

[View on casel.org](http://casel.org)

Michigan K-12 Math Standards

The Michigan Department of Education's full K-12 Standards for Mathematics document.

[View Michigan DOE PDF](#)

References

- Black, P., & Wiliam, D. (1998). Inside the black box: Raising standards through classroom assessment. *Phi Delta Kappan*, 80(2), 139-148.
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